Shreyash Katiyar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.08
Dr Virendra Swaroop Education Center, Kanpur.	Class XII (ICSE)	2024	9.18%
Dr Virendra Swaroop Education Center, Kanpur.	Class X (ICSE)	2022	9.46%

EXPERIENCE AND CERTIFICATIONS

Machine Learning Intern / Mentee *Launchpad & BTech Waley* | June 2025 - *July 2025 (Certificate)*

- Completed an intensive, month-long global mentorship program focused on applied Machine Learning concepts.
- Cleaned and pre-processed datasets using Python, pandas, and NumPy to prepare for model training.
- Conducted Exploratory Data Analysis (EDA) and applied feature engineering techniques to optimize data for clustering algorithms like K-means and PCA.
- Developed and evaluated predictive models using regression and classification techniques, improving accuracy by up to 95%

ATF 2025 Stage-1 Qualifying Candidate (Certificate)

- Qualified Stage-1 of the AlgoUniversity Tech Fellowship (ATF) 2025, ranking in the top 8% nationwide out of 250,000 participants.

SKILLS

Languages: C, C++, MySQL, Python, HTML, CSS, JavaScript.

Tools & Libraries: Git/GitHub, LaTeX, Jupterlab/Google Colab, TensorFlow/Keras, VS-Code, Pandas, NumPy, Scikit-learn.

Developer Profiles: GitHub-<link> , Codeforces-<link>

RELEVANT COURSES

Computer Science: Introduction to Programming , Data Structutes, Algorithms, Theory of Computation, DBMS-1, Compilers-1.

Applied Mathematics: Linear Algebra, Applied Statistics, Convex Optimization.

Pure Mathematics: Calculus, Number Theory, Ordinary Differential Equations, Transform Techniques, Group Theory.

PROJECTS

Vantage Point Tree (VPT) Plagiarism Detector | (GitHub Link)

C++, Git

- **Engineered a high-performance document similarity search engine** utilizing a custom Vantage Point Tree, optimizing spatial search time complexity from O(n) to O(log n).
- **Developed a custom text parsing pipeline** to extract alphabetical character frequencies, applying L2 normalization to generate robust 26-dimensional feature vectors.
- **Implemented spatial partitioning and Cosine Distance metrics** to recursively prune the multi-dimensional search

space via the triangle inequality, enabling rapid nearest-neighbor document retrieval.

2D Physics Game Engine (Asteroids) | (GitHub Link)

C++, SDL2, Make

- **Developed a real-time 2D game engine** from scratch using C++ and the SDL2 library for hardware-accelerated rendering and low-level input event handling.
- **Engineered frame-rate independent physics** by implementing a custom delta-time game loop and robust 2D vector mathematics to accurately calculate trajectory, rotational momentum, and dynamic ship thrust.
- **Managed dynamic memory and entity lifecycles** by utilizing Standard Template Library (STL) containers to efficiently spawn, track, and cull hundreds of on-screen game objects during real-time collision detection.

AI-Powered Image Colorizer | TensorFlow, Keras, Python, OpenCV | <GitHub Link>

- Developed a convolutional neural network (CNN) to automatically colorize grayscale images by predicting chrominance channels in the LAB color space.
- Built an end-to-end deep learning pipeline including image preprocessing, model training, inference, and visualization using TensorFlow/Keras and OpenCV.
- Implemented data preprocessing, model checkpointing, and early stopping to improve training efficiency and model performance.
- Designed the system to generate realistic colorized outputs from unseen grayscale images, demonstrating practical computer vision and deep learning skills.